

RECON WORKSHEET

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Target: google.com
Date: 4 September 2026
Authorization/source: Public training target. Only passive lookups were used.
Nothing was scanned, logged into or attacked.

Tools used: whois, dig, nslookup, curl, traceroute

1. WHOIS

Registrar: MarkMonitor, Inc.
Creation date: 15 September 1997
Expiration date: 14 September 2028
Nameservers: ns1.google.com
ns2.google.com
ns3.google.com
ns4.google.com
Domain status: clientDeleteProhibited
clientTransferProhibited
clientUpdateProhibited
serverDeleteProhibited
serverTransferProhibited
serverUpdateProhibited

These are locks. They stop the domain being moved, changed or deleted unless the locks are lifted first. All six are set, which is the safest setup.

DNSSEC: unsigned
Registrant: Google LLC, United States
The contact email is hidden behind a web form.

2. DNS

Resolver used: 192.168.1.1 (the local router)
The answers were marked non-authoritative, so they came from a cache and not from Google's own nameservers.

A: 142.251.39.206
AAAA: 2a00:1450:4007:809::200e
NS: ns1.google.com, ns2.google.com,
ns3.google.com, ns4.google.com
MX: 10 smtp.google.com

TXT: 17 records in total.

SPF record:
v=spf1 include:_spf.google.com ~all

14 verification records. These are strings a company adds to prove it owns the domain when signing up to an outside service. The ones here name Google, Microsoft, Facebook, Apple, DocuSign, OneTrust, Cisco, GlobalSign and Arcules.

2 records with no obvious label.

DMARC: v=DMARC1; p=reject; rua=mailto:mailauth-reports@google.com
(checked separately, it was not in the saved dig output)

3. HTTP

Command used: curl -I https://www.google.com

Status: HTTP/2 200
Server: gws (no version number given)
Content-Type: text/html; charset=ISO-8859-1
Cache-Control: private
HSTS: not present

Other security headers:

X-Frame-Options: SAMEORIGIN	present
X-XSS-Protection: 0	present
Content-Security-Policy	Report-Only
X-Content-Type-Options	missing
Referrer-Policy	missing
Permissions-Policy	missing

Also seen:

alt-svc: h3=":443", so HTTP/3 is supported
Two cookies were set (AEC and __Secure-ENID). Both have Secure, HttpOnly and SameSite=lax.

4. NETWORK PATH

Traceroute tool: traceroute on Linux
Approximate hops: 6
Notable hops: 1 100.64.5.91, which is carrier NAT
2 datapacket.com
3 cdn77.com in Zurich
4 google-zur.cdn77.com
6 destination reached
Timeouts: hop 5 gave no reply
Observed providers: DataPacket, CDN77, Google

Note on this section:

The traceroute file saved with this task was empty, so no route was captured the first time round. I ran it again afterwards on a different machine. That machine resolved google.com to a different address than dig had earlier, so the route above is not the route from the original host. This section should be run again on the machine that produced the dig output:

```
traceroute google.com | tee "traceroute output"
```

5. SUMMARY

DNS observations:

DNSSEC is not turned on. Without it a resolver cannot check that the answers it receives are genuine. That is not something an attacker can use on its own, but it takes away one layer of protection.

The TXT records give away part of the supplier list. Fourteen of them name an outside service the company has signed up to. Anyone can read these, and in a real assessment they are a good starting point for working out what else is worth looking at.

SPF ends in ~all, which is a soft fail. A hard fail (-all) would be stricter. DMARC makes up for it though, as it is set to p=reject.

There is one MX record and all four nameservers belong to the same company. That works fine day to day but leaves no second provider to fall back on.

All the answers came out of the router's cache rather than from Google's nameservers directly, so they show what was cached at the time.

IP observations:

142.251.39.206 sits inside the range 142.250.0.0 to 142.251.255.255, which is registered to Google LLC in the United States. The address is Google's own rather than a rented server or an outside CDN.

The domain answers on both IPv4 and IPv6. This is easy to forget, and firewall rules and scan scopes are often written for IPv4 only and miss the IPv6 side completely.

The TTL on the A record was very short, so the address changes often. The one recorded here is only one of many.

HTTP observations:

No HSTS header came back. On most targets a missing HSTS header is a low severity finding and would go in the report.

X-Content-Type-Options, Referrer-Policy and Permissions-Policy are all missing as well.

There is a Content-Security-Policy, but it is set to Report-Only. That means the browser reports problems back but does not actually block anything.

X-Frame-Options is set to SAMEORIGIN, which helps against clickjacking.

The Server header only says gws with no version number. Hiding the version is good practice as it gives less away about what is running.

X-XSS-Protection is set to 0. This looks wrong at first but it is deliberate and correct. The old browser XSS filter caused problems of its own and modern browsers have dropped it. Scanners often flag this, and they are wrong to.

Two cookies were set before logging into anything, but both have the right flags on them.

Network observations:

The route goes out through carrier NAT, then DataPacket, then CDN77 in Zurich before reaching Google.

Hop 5 gave no reply. This is normal. Plenty of routers are set not to answer traceroute, and a silent hop does not mean something is being blocked.

Traceroute only shows one route out of several possible ones, and the traffic may well take a different path next time.

The trace itself has a problem with it, explained in the note in section 4.

6. LIMITATIONS

- WHOIS information may be privacy-protected.
- DNS may point to shared/CDN/cloud infrastructure.
- Traceroute does not necessarily reveal the complete physical path.
- HTTP headers can change over time.
- The DNS answers came from a cache, not from the nameservers themselves.
- The traceroute was run from a different machine and needs running again.
- This is one snapshot of a large site that is spread across many servers. Someone running the same commands in another country would get different addresses and possibly different headers.
- Passive lookups only. No port scanning or vulnerability testing was done, so nothing here says anything about what services are running or how they behave.